Answer Key & Teacher Notes

Session 1 — Equations and Solutions · Session 2 — Writing Equations

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Equations and Solutions

TALK PROMPTS

- Why can an equation be false for one value of the variable and true for another?
- Two partners tested different values and got different results. How can both checks still be correct work?

PARTNER TASK

Form A, problem 1a. The equation is s − 15 = 25. One partner tests s = 40; the other tests s = 35. Each shows the substitution, then together they decide which value is the solution and why.

WATCH FOR

- Reporting the value of each side (42) as the solution instead of the value of the variable (12).
- Reading 4p as 4 + p instead of 4 × p.
- Accepting a value that makes the two sides "close" but not equal.

SESSION 2 · TEACHER NOTES

Writing Equations

TALK PROMPTS

- Why is a good estimate a better starting point than a random number?
- Why does your first estimate not have to be right for this method to work?

PARTNER TASK

Coat Drive. A drive collected c coats on Friday and 37 more on Saturday, 82 in all. One partner writes the equation; the other estimates and tests one value. Switch roles and revise until both sides match.

WATCH FOR

- Writing the parts in the wrong order for a take-away story (72 + m = 25 or 18 − b = 60).
- Writing 9 − n for "a number decreased by 9."
- Using + for an equal-groups story instead of ×.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.EE.B.5	1	2	2	1	1	2	2	3	2	1	1
S1 · Form B	6.EE.B.5	1	2	2	1	1	2	2	3	2	1	1
S2 · Form A	6.EE.B.6	1	2	2	2	2	2	2	3	2	1	1
S2 · Form B	6.EE.B.6	1	2	2	2	2	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

equation: A math sentence that says two amounts are **equal**.

variable: A letter that stands for an **unknown** number.

solution: A value that makes the equation **true**.

substitute: **Replace** the variable with a number.

- To test a value, **substitute** it for the variable and do the math.
- If both sides are **equal**, the value is a solution. If not, it is not a solution.

Example: $k = 7$: $7 - 4 = \mathbf{3}$. $3 \neq 2$, so the equation is **false**. $k = 6$: $6 - 4 = \mathbf{2}$. $2 = 2$, so the equation is true. Answer: $k = \mathbf{6}$ is the solution.

YOU TRY

1. a. T (20 = 20) b. F (2 ≠ 6)

2. m = 8, because 7 × 8 = 56

EXIT TICKET

1. a. T (19 = 19) b. F (18 ≠ 21)

2. h = 8, because 8 − 5 = 3

3. **A** (value)

PRACTICE FORM A

1. a. T (25 = 25) b. F (35 ≠ 42) c. T (47 = 47) d. F (128 ≠ 36) e. T (3 = 3) f. T (41 = 41)

2. a. 21 + 12 = 33, true b. 19 + 12 = 31 ≠ 33, false c. j = 21

3. p = 12, because 6 × 12 = 72

4. 60 − 24 = 36, so yes, c = 60 is a solution.

5. 4 × 7 = 28, so yes, Ray is correct.

6. f − 2 = 11; f = 13, because 13 − 2 = 11

7. 5 × 8 = 40 ≠ 45, so g = 8 is not a solution. g = 9, because 5 × 9 = 45.

8. a. n + 17 = 45 b. 28 + 17 = 45, true, so n = 28 is the solution c. Testing checks one given value; solving finds the value that makes the equation true.

9. Her check is right, but the solution is the value of the variable that makes the equation true. The solution is k = 12; 42 is the value of each side.

10. **A** (yes)

11. **B** (no)

Self-check word **EQUAL**

PRACTICE FORM B

1. a. T (30 = 30) b. F (54 ≠ 48) c. T (51 = 51) d. F (72 ≠ 27) e. T (4 = 4) f. T (44 = 44)

2. a. 27 + 14 = 41, true b. 25 + 14 = 39 ≠ 41, false c. j = 27

3. p = 13, because 8 × 13 = 104

4. 65 − 18 = 47, so yes, c = 65 is a solution.

5. 5 × 6 = 30 ≠ 35, so no, Ray is not correct (t = 7).

6. f − 3 = 14; f = 17, because 17 − 3 = 14

7. 7 × 9 = 63 ≠ 56, so g = 9 is not a solution. g = 8, because 7 × 8 = 56.

8. a. n + 19 = 52 b. 33 + 19 = 52, true, so n = 33 is the solution c. Testing checks one given value; solving finds the value that makes the equation true.

9. 40 ≠ 45, so x = 15 is not a solution. Both sides must be exactly equal. The solution is x = 20, because 20 + 25 = 45.

10. **A** (yes)

11. **B** (no)

Self-check word **VALUE**

GUIDED NOTES

variable: A letter that stands for the **unknown** number.

equation: A math sentence with an **equal** sign.

increased by: Means **add**.

decreased by: Means **subtract**.

- Name the unknown with a **letter** and say what it means.
- Write the story as a math sentence, like start **–** lost = left, or groups × size = **total**.

Example: Let m = the marbles that **rolled away**. Start − lost = left: $30 - m =$ **11**. Test $m = 19$: $30 - 19 =$ **11**. True. Answer: **19** marbles rolled away.

YOU TRY

1. a. $n + 8 = 15$ b. $3n = 21$

2. $46 - c = 13$; $c = 33$ cartons ($46 - 33 = 13$)

EXIT TICKET

1. a. $n + 14 = 31$ b. $4n = 36$

2. $50 - b = 32$; $b = 18$ books

3. **B** ($8w = 56$)

PRACTICE FORM A

1. a. $n + 17 = 40$ b. $6n = 54$ c. $n - 8 = 19$ d. $n \div 3 = 7$

2. a. $s + 28 = 61$ b. $s = 33$, because $33 + 28 = 61$

3. a. $m - 15 = 39$ b. $50 - 15 = 35 \neq 39$, too small; $m = 54$, because $54 - 15 = 39$

4. $44 - g = 26$; $g = 18$ students

5. $58 + w = 90$; $w = 32$ liters

6. $c - 75 = 22$; $c = 97$ cans, because $97 - 75 = 22$

7. a. $45b = 180$ b. $45 \times 4 = 180$, true, so $b = 4$ buses

8. a. Let s = the number of students; $120 + 8s = 360$ b. $120 + 8 \times 30 = 120 + 240 = 360$, so yes c. $8s = 360$ ($s = 45$)

9. Removing means subtract, so the equation should be $72 - m = 25$. His answer $m = 47$ is right: $72 - 47 = 25$. ($72 + 47$ is not 25.)

10. **B** ($n + 12 = 30$)

11. **D** ($n - 9 = 14$)

Self-check word **TOTAL**

PRACTICE FORM B

1. a. $n + 23 = 50$ b. $8n = 72$ c. $n - 11 = 25$ d. $n \div 6 = 5$

2. a. $c + 37 = 82$ b. $c = 45$, because $45 + 37 = 82$

3. a. $m - 12 = 44$ b. $50 - 12 = 38 \neq 44$, too small; $m = 56$, because $56 - 12 = 44$

4. $52 - g = 35$; $g = 17$ students

5. $47 + w = 80$; $w = 33$ liters

6. $c - 64 = 29$; $c = 93$ cans, because $93 - 64 = 29$

7. a. $40b = 160$ b. $40 \times 4 = 160$, true, so $b = 4$ buses

8. a. Let s = the number of students; $90 + 6s = 270$ b. $90 + 6 \times 30 = 90 + 180 = 270$, so yes c. $6s = 270$ ($s = 45$)

9. The starting amount comes first: $60 - b = 18$. Her answer $b = 42$ is right: $60 - 42 = 18$. ($18 - 42$ is not 60.)

10. **C** ($7n = 63$)

11. **D** ($n - 6 = 21$)

Self-check word **SOLVE**

SHEET 1 · EQUATIONS AND SOLUTIONS

1. a. T (20 = 20) b. F (30 ≠ 35) c. T (16 = 16) d. F (8 ≠ 7) e. T (30 = 30) f. T (54 = 54) g. T (4 = 4) h. F (72 ≠ 63)

2. x = 25. Checks: 15 + 16 = 31 (no); 25 + 16 = 41 (yes); 35 + 16 = 51 (no).

3. b = 11: 11 − 7 = 4, and 4 ≠ 18, so no. b = 25: 25 − 7 = 18, so yes. The solution is b = 25.

4. 4p means 4 × p, not 4 + p. Substituting gives 4 × 8 = 32, and 32 ≠ 12, so p = 8 is not a solution. The solution is p = 3, because 4 × 3 = 12.

5. t = 36: 6 × 36 = 216, and 216 ≠ 42, so no. t = 7: 6 × 7 = 42, so yes. The solution is t = 7 tickets.

SHEET 2 · WRITING EQUATIONS

1. a. n + 9 = 31 b. 7n = 63 c. n − 12 = 20 d. n ÷ 5 = 8

2. s − 14 = 29. Test: 43 − 14 = 29, so s = 43 is the solution.

3. 4v = 56. Test 12: 4 × 12 = 48 (no). Test 14: 4 × 14 = 56 (yes). v = 14 students.

4. 5p = 85. Test: 5 × 17 = 85, so p = 17 pages a day is the solution.

5. Omar's equation gives b = 17, which would make the brother older than Tia. The brother's age plus 4 is Tia's age: b + 4 = 13, so b = 9.

6. **B** ($9w = 72$)

7. **C** ($n = 42$)

October 2026 conceptual additions — teacher key

Session 1 · Equations and Solutions

6−3=3, not 4; 7−3=4 is true, so x=7 is a solution.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: 6−3=3, not 4; 7−3=4 is true, so x=7 is a solution.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Writing Equations

r is removed counters; 28−r=9; r=19. Check 28−19=9. An equivalent equation r+9=28 is valid.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: r is removed counters; 28−r=9; r=19. Check 28−19=9. An equivalent equation r+9=28 is valid.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.